

Lecture 10

Randomized Complete Block Designs, Factorial Designs, and Split-Plot Designs

Reading: Oehlert 2010 Chapters 8; 13.1-13.2; 16.1-16.3; DAE
2017 Chapters 6.1-6.5; 10; 19.1-19.3

STAT 8020 Statistical Methods II

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Agenda

Randomized
Complete Block
Designs, Factorial
Designs, and
Split-Plot Designs



Randomized Complete
Block Designs

Factorial Designs

Split-Plot Designs

1 Randomized Complete Block Designs

2 Factorial Designs

3 Split-Plot Designs

Review: Completely Randomized Design (CRD)

Randomized
Complete Block
Designs, Factorial
Designs, and
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A Completely Randomized Design (CRD):

- Uses g treatment levels
- **Randomly** assigns **treatments** to **experimental units**
- Assumes experimental units are approximately homogeneous

Randomized Complete
Block Designs

Factorial Designs

Split-Plot Designs

Key Assumption

All experimental units are similar enough that observed differences are primarily due to treatments.

Review: Completely Randomized Design (CRD)

A Completely Randomized Design (CRD):

- Uses g treatment levels
- **Randomly** assigns **treatments** to **experimental units**
- Assumes experimental units are approximately homogeneous

Key Assumption

All experimental units are similar enough that observed differences are primarily due to treatments.

Question: What happens if experimental units are *not* homogeneous?

Randomized Complete Block Design (RCBD)

Idea: Group similar experimental units together.

- Divide experimental units into relatively homogeneous groups called **blocks**
- **Randomize** treatments *within each block*
- Each treatment usually appears once per block

Goal of Blocking

Reduce unwanted variability so treatment comparisons become more precise.

Key Assumption:

Treatment effects are consistent across blocks $\Rightarrow$ no treatment-by-block interaction (**additive assumption**)

RCBD Notation

- g : number of treatments
- r : number of blocks
- $N = rg$: total number of observations
- y_{ij} : observation from block i receiving treatment j

Blocks	Treatments				
	1	2	... j	j	g
1	y_{11}	y_{12}	...	y_{1j}	y_{1g}
2	y_{21}	y_{22}	...	y_{2j}	y_{2g}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i	y_{i1}	y_{i2}	...	y_{ij}	y_{ig}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
r	y_{r1}	y_{r2}	...	y_{rj}	y_{rg}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
r	y_{r1}	y_{r2}	...	y_{rj}	y_{rg}

$$\bar{y}_{\cdot j} = \frac{1}{r} \sum_{i=1}^r y_{ij}$$

Treatment mean for treatment j

$$\bar{y}_{i\cdot} = \frac{1}{g} \sum_{j=1}^g y_{ij}$$

Block mean for block i

$$\bar{y}_{\cdot\cdot} = \frac{1}{N} \sum_{i=1}^r \sum_{j=1}^g y_{ij}$$

Overall mean

The RCBD model is

$$y_{ij} = \mu + \alpha_j + \beta_i + \varepsilon_{ij}, \quad i = 1, \dots, r, \quad j = 1, \dots, g,$$

where

- μ : baseline mean
- α_j : treatment effect
- β_i : block effect
- $\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$

Important Assumption

No treatment-by-block interaction exists.
Treatment effects are assumed to be additive across blocks.

Partitioning Variability in RCBD

Total variability is divided into:

$$SS_{tot} = SS_{trt} + SS_{blk} + SS_{err}$$

- Treatment SS: variation due to treatments

$$SS_{trt} = \sum_{j=1}^g r(\bar{y}_{.j} - \bar{y}_{..})^2$$

- Block SS: variation explained by blocks

$$SS_{blk} = \sum_{i=1}^r g(\bar{y}_{i.} - \bar{y}_{..})^2$$

- Error SS: unexplained variation

$$SS_{err} = \sum_{i=1}^r \sum_{j=1}^g (y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..})^2$$

Blocking reduces the error variation $\Rightarrow$ more powerful tests

ANOVA Table and F Test

Source	df	SS	MS	F statistic
Treatment	$g - 1$	SS_{trt}	$MS_{trt} = \frac{SS_{trt}}{g-1}$	$F_{trt} = \frac{MS_{trt}}{MS_{err}}$
Block	$r - 1$	SS_{blk}	$MS_{blk} = \frac{SS_{blk}}{r-1}$	$F_{blk} = \frac{MS_{blk}}{MS_{err}}$
Error	$(g - 1)(r - 1)$	SS_{err}	$MS_{err} = \frac{SS_{err}}{(g-1)(r-1)}$	
Total	$N - 1$	SS_{tot}		

There are two hypothesis tests in an RCBD:

- $H_0 : \alpha_j = 0 \quad j = 1, \dots, g$
 $H_a : \alpha_j \neq 0 \quad \text{for some } j$

Test Statistic: $F_{trt} = \frac{MS_{trt}}{MS_{err}}$. Under H_0 ,
 $F_{trt} \sim F_{df_1=g-1, df_2=(g-1)(r-1)}$

- H_0 : The means of all blocks are equal
 H_a : At least one of the blocks has a different mean

Test Statistic: $F_{blk} = \frac{MS_{blk}}{MS_{err}}$. Under H_0 ,
 $F_{blk} \sim F_{df_1=r-1, df_2=(g-1)(r-1)}$

Concrete Drying Study: Why Blocking Matters

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Split-Plot Designs

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Randomized Complete
Block Designs

Factorial Designs

Split-Plot Designs

Concrete cylinders for bridge supports are produced using three drying methods: **A, B, and C**.

Goal: determine which method gives the highest compressive strength.



Each batch produces exactly three cylinders, but concrete quality may vary substantially from batch to batch

Blocking: Use each batch as a block: assign A, B, and C within every batch.

Data: $r = 5$ batches and $g = 3$ drying methods. The response is cylinder compressive strength from a destructive test

Example: Data Set

The data are:

Treatment	Batch					Trt Sum
	1	2	3	4	5	
A	52	47	44	51	42	236
B	60	55	49	52	43	259
C	56	48	45	44	38	231
Batch Mean	168	150	138	147	123	726

The primary null hypothesis is that all three drying techniques are equivalent, in terms of compressive strength.

The secondary null is that the batches are equivalent (but if they are, then we have wasted power by controlling for an effect that is small or non-existent).

Example: RCBD ANOVA Interpretation

Analysis of Variance Table

Response: x

	Df	Sum Sq	Mean Sq	F value
trt	2	89.2	44.60	7.6239
blk	4	363.6	90.90	15.5385
Residuals	8	46.8	5.85	
Pr(>F)				
trt		0.0140226	*	
blk		0.0007684	***	

- The treatment effect is statistically significant ($p = 0.014$)
- The block effect is also significant ($p < 0.001$) $\Rightarrow$ batch-to-batch variability is substantial
- Blocking successfully removed important nuisance variability

Key Takeaway

Accounting for block effects increases the sensitivity of treatment comparisons.

What If We Ignore the Block Effect?

If we ignore the batch effect:

Treatment		Trt Sum
A	52, 47, 44, 51, 42	236
B	60, 55, 49, 52, 43	259
C	56, 48, 45, 44, 38	231

- Residual variability becomes much larger
- Treatment differences become harder to detect
- Statistical power decreases

Important Lesson

Good experimental design can dramatically improve inference without collecting more data

Analysis of Variance Table

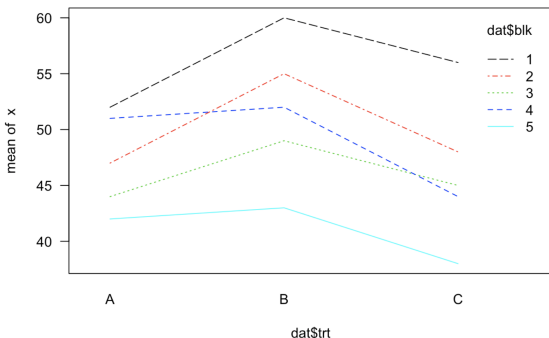
Response: x

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
trt	2	89.2	44.6	1.3041	0.3073
Residuals	12	410.4	34.2		

We **fail to reject** the null $H_0 : \mu_A = \mu_B = \mu_C$ if we **ignore the block effect**

⇒ Using blocks in this example gave us a more powerful test!

Assessing the Additivity Assumption: Interaction Plot



- Approximately parallel lines support the RCBD assumption that treatment effects are consistent across blocks
- Different slopes or crossing lines suggest a possible treatment-by-block interaction
- Strong interaction may reduce the effectiveness of blocking and violate the standard RCBD model assumptions

The Battery Design Experiment (Example 5.1, Montgomery, 6th Ed)

An engineer studies how **material type** and **temperature** affect battery life using three materials and three temperature levels:

Material Type	Temperature (°F)					
	15		70		125	
1	130	155	34	40	20	70
	74	180	80	75	82	58
2	150	188	136	122	25	70
	159	126	106	115	58	45
3	138	110	174	120	96	104
	168	160	150	139	82	60

This design is called a **3^2 factorial design**

Why Factorial Designs?

Factorial designs allow us to study:

- Multiple factors simultaneously
- Main effects
- Interactions between factors

The **effects model**:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk},$$

$$i = 1, \dots, a, j = 1, \dots, b, k = 1, \dots, n$$

- a and b : the numbers of levels in factors A and B
- $(\alpha\beta)_{ij}$: the interaction between α_i and $\beta_j \Rightarrow$ measures whether the effect of one factor depends on the level of another factor
- **Constraints:**
$$\sum_{i=1}^a \alpha_i = \sum_{j=1}^b \beta_j = \sum_{i=1}^a (\alpha\beta)_{ij} = \sum_{j=1}^b (\alpha\beta)_{ij} = 0$$
- abn : total number of observations; **replication** (n) allows testing **interaction** effects

ANOVA Table

Source	df	SS	MS	F
Factor A	$a - 1$	SS_A	$MS_A = \frac{SS_A}{a-1}$	$F = \frac{MS_A}{MS_E}$
Factor B	$b - 1$	SS_B	$MS_B = \frac{SS_B}{b-1}$	$F = \frac{MS_B}{MS_E}$
Interaction	$(a - 1)(b - 1)$	SS_{AB}	$MS_{AB} = \frac{SS_{AB}}{(a-1)(b-1)}$	$F = \frac{MS_{AB}}{MS_E}$
Error	$ab(n - 1)$	SS_E	$MS_E = \frac{SS_E}{ab(n-1)}$	
Total	$abn - 1$	SS_T		

```
lm <- lm(y ~ temp * material, data = dat)
anova(lm)
```

```

### Analysis of Variance Table

Response: y

|               | Df | Sum Sq | Mean Sq | F value | Pr(>F)        |
|---------------|----|--------|---------|---------|---------------|
| temp          | 2  | 39119  | 19559.4 | 28.9677 | 1.909e-07 *** |
| material      | 2  | 10684  | 5341.9  | 7.9114  | 0.001976 **   |
| temp:material | 4  | 9614   | 2403.4  | 3.5595  | 0.018611 *    |
| Residuals     | 27 | 18231  | 675.2   |         |               |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Randomized  
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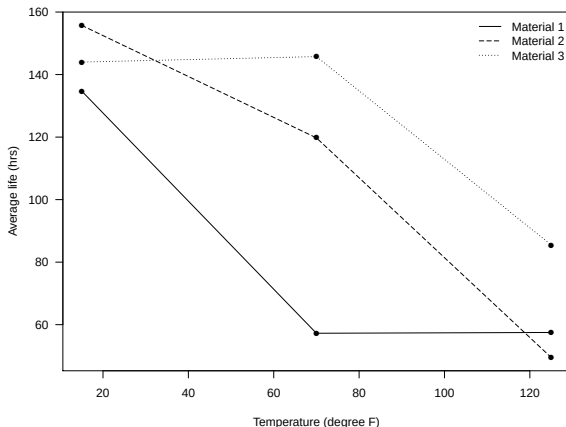
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Randomized Complete  
Block Designs

Factorial Designs

Split-Plot Designs

# Interaction Plot



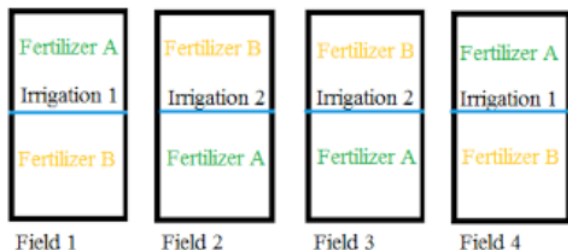
## Interpretation

The lines are not parallel, suggesting a possible interaction between temperature and material type  $\Rightarrow$  the effect of temperature depends on the material used

## An Example of Split-Plot Design

Suppose we wish to determine the effects of two fertilizers (“A” and “B”) and two irrigation methods (“1” and “2”) on yield

- **Treatments:** Fertilizer types and irrigation methods
- **Experimental Unit:** a small region



### Motivation

Split-plot designs arise when some factors are difficult or expensive to randomize completely

$$y_{ijk} = \mu + \alpha_i + \delta_{k(i)} + \beta_j + \alpha\beta_{ij} + \varepsilon_{k(ij)},$$

$$i = 1, \dots, a, j = 1, \dots, b, k = 1, \dots, n$$

- $\mu$ : overall mean
- $\alpha_i$ : effect of  $i_{\text{th}}$  level of the whole-plot factor
- $\beta_j$ : effect  $j_{\text{th}}$  split-plot factor
- $(\alpha\beta)_{ij}$ : joint effect of  $i_{\text{th}}$  level of the whole-plot factor and  $j_{\text{th}}$  split-plot factor
- $\delta_{k(i)} \sim N(0, \sigma_\delta^2)$  and  $\varepsilon_{k(ij)} \sim N(0, \sigma_\varepsilon^2)$ : the whole-plot and the split-plot level random errors

$$y_{ijk} = \mu + \alpha_i + \delta_{k(i)} + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{k(ij)}$$

$$i = 1, \dots, a, \quad j = 1, \dots, b, \quad k = 1, \dots, n$$

- $\mu$ : baseline mean;  $\alpha_i$ : whole-plot factor effect  
 $\beta_j$ : split-plot factor effect;  $(\alpha\beta)_{ij}$ : interaction effect
- $\delta_{k(i)} \sim N(0, \sigma_\delta^2)$ : whole-plot error  
 $\varepsilon_{k(ij)} \sim N(0, \sigma_\varepsilon^2)$ : split-plot error

## Key Idea in Split-Plot Designs

- Whole-plot factors use larger experimental units
- Split-plot factors use smaller sub-units
- Therefore, two different error terms are needed

## ANOVA Table

Randomized  
Complete Block  
Designs, Factorial  
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Randomized Complete  
Block Designs

Factorial Designs

Split-Plot Designs

| Source      | df                | SS         | MS                                        | F                              |
|-------------|-------------------|------------|-------------------------------------------|--------------------------------|
| Factor A    | $a - 1$           | $SS_A$     | $MS_A = \frac{SS_A}{a-1}$                 | $F = \frac{MS_A}{MS_{E_1}}$    |
| Error 1     | $a(n - 1)$        | $SS_{E_1}$ | $MS_{E_1} = \frac{SS_{E_1}}{a(n-1)}$      |                                |
| Factor B    | $b - 1$           | $SS_B$     | $MS_B = \frac{SS_B}{b-1}$                 | $F = \frac{MS_B}{MS_{E_2}}$    |
| Interaction | $(a - 1)(b - 1)$  | $SS_{AB}$  | $MS_{AB} = \frac{SS_{AB}}{(a-1)(b-1)}$    | $F = \frac{MS_{AB}}{MS_{E_2}}$ |
| Error 2     | $a(b - 1)(n - 1)$ | $SS_{E_2}$ | $MS_{E_2} = \frac{SS_{E_2}}{a(b-1)(n-1)}$ |                                |
| Total       | $abn - 1$         | $SS_T$     |                                           |                                |

The  $F$ -tests use different denominators:

Factor A  $\rightarrow MS_{E_1}$ ,

Factor B and Interaction  $\rightarrow MS_{E_2}$



## Split-Plot Example: Fertilizer and Strawberry Varieties

Farmer John has eight plots of land. Two fertilization schemes (control and new) are randomly assigned to the plots (the **whole-plots**).

Each whole-plot is divided into four subplots, and four strawberry varieties are randomly assigned within each plot (the **split-plots**).

**Goal:** study the effects of fertilization scheme and strawberry variety on fruit mass

### R output:

```
library(lmerTest)
fit <- lmer(mass ~ fertilizer * variety + (1 | plot), data = dat)
anova(fit)

Type III Analysis of Variance Table with Satterthwaite's method
##
```

|                       | Sum Sq  | Mean Sq | NumDF | DenDF | F value | Pr(>F)        |
|-----------------------|---------|---------|-------|-------|---------|---------------|
| ## fertilizer         | 137.413 | 137.413 | 1     | 6     | 68.2395 | 0.0001702 *** |
| ## variety            | 96.431  | 32.144  | 3     | 18    | 15.9627 | 2.594e-05 *** |
| ## fertilizer:variety | 4.173   | 1.391   | 3     | 18    | 0.6907  | 0.5695061     |

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The R model syntax may look unfamiliar because split-plot designs are typically analyzed using **mixed-effects models**, which will be covered in the next module

## Summary

These slides cover:

- Randomized Complete Block Designs (RCBD)
- Factorial Designs
- Split-Plot Designs

R functions to know:

- **Visualize interaction:** Utilize `interaction.plot`
- **Split-plot design analysis:** Employ `lmer` from the package `lmerTest` to fit linear mixed-effects models